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$$\mu = \frac{1}{n \cdot h \cdot w} \sum_{k=1}^{n} \sum_{i=1}^{h} \sum_{j=1}^{w} p_{ij}^{(k)}$$ ⇒ mean across pixels of an image

$p_{ij}^{(k)}$: pixel at (i,j) of image I_k

$$\mu_{ij} = \frac{1}{n} \sum_{k=1}^{n} p_{ij}^{(k)}$$ is the pixel-wise mean across all imgs.

L1 dist between test and train image:

1

$$\sum \sum | p_{test} - p_{train} |$$

Subtract mean: $\tilde{p}_{ij}^{(k)} = p_{ij}^{(k)} - \mu$

$$L_2 = \sum \sum |(p_{test} - \mu) - (p_{train} - \mu)|$$

$$= \sum \sum | p_{test} - \mu - p_{train} + \mu |$$

$$= \sum \sum | p_{test} - p_{train} |$$

⇒ Perf does not change.

② Pixel-wise mean across images:

$$\mu_{ij} = \frac{1}{n}\sum_{k=1}^{n} p_{ij}^{(k)}$$

$$\tilde{p}_{ij}^{(k)} = p_{ij}^{(k)} - \mu_{ij} = p_{ij}^{(k)} - \frac{1}{n}\sum_{k=1}^{n} p_{ij}^{(k)}$$

At pixel $(i; j)$

$$\tilde{p}_{ij}^{(test)} = p_{ij}^{test} - \frac{1}{n}\sum_{k=1}^{n} p_{ij}^{(k)}$$

$$\tilde{p}_{ij}^{(train)} = p_{ij}^{(test)} - \frac{1}{n}\sum_{k=1}^{n} p_{ij}^{(k)}$$

$$\Rightarrow \sum\sum \left| \tilde{p}_{ij}^{(test)} - \tilde{p}_{ij}^{(train)} \right|$$

$$= \sum\sum \left| p_{ij}^{(test)} - p_{ij}^{(train)} \right|$$

$\Rightarrow$ Perf does not change.

③ Same for dividing by SD